

LE VAN LONG

Hanoi, Vietnam

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Education

Hanoi University of Mining and Geology (HUMG)

(Information Technology)

- Degree Classification: Very Good
- GPA current: 3.22/4.00

Hanoi, Vietnam

Aug 2022 – Present

Projects

AnKhang Assistant

• Project Description:

- **Technologies:** LangChain, LangGraph, Groq, Qdrant, Redis, ChromaDB, FastAPI, Streamlit, Selenium, Docker
- Developed an agent-based RAG chatbot leveraging LangGraph for workflow orchestration and WebSocket to enable real-time customer interaction. The system delivers responses within 2–4 seconds for straightforward queries and 7–10 seconds for more complex ones.
- Enhanced answer quality through techniques such as query rewriting, context-aware query routing using conversation history, and few-shot prompting..
- Using hybrid search by combining sparse retriever (BM25), dense retriever, and Maximal Marginal Relevance (MMR) method to retrieve information, reranking techniques to improve Top-K retrieval relevance
- Implemented a hybrid retrieval strategy by integrating BM25 (sparse retrieval), dense embeddings, and Maximal Marginal Relevance (MMR), along with reranking methods to improve Top-K result relevance.
- Gathered domain-specific knowledge by building a web crawling pipeline using Selenium.
- link : https://github.com/longlee04/AI_Asistant-

Titanic Survivor Prediction

• Project Description:

- Conducted a comparative study of various machine learning algorithms on the Titanic dataset from Kaggle to identify the most effective predictive model.
- Carried out comprehensive data preprocessing, including missing value imputation, feature normalization, and exploratory data analysis (EDA) to uncover key patterns. Applied dimensionality reduction techniques such as Principal Component Analysis (PCA) to optimize feature space.
- Built and trained multiple models using Scikit-Learn, including Logistic Regression, Decision Tree, Random Forest, Support Vector Machine (SVM), and XGBoost. The SVM model delivered the best performance, achieving a Kaggle score of 0.78.
- Assessed model effectiveness using evaluation metrics such as confusion matrix, accuracy, and F1-score to ensure reliable and balanced predictions.

Skills & Interests

- Java, Python, C/C++ , SQL
- Scikit-learn, Langchain, Langgraph, Streamlit